

Stable 2-Cycles in the Iteration of the Principal Negative Complex Root: An Analysis of $z_{k+1} = (-z_k)^{1/n}$

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Abstract

The iterated sequence $z_{k+1} = f(z_k)$ in the complex plane provides a rich field of study in dynamical systems. A subtle change, such as introducing a negation, can dramatically alter the system's long-term behavior. This paper analyzes the recurrence $z_{k+1} = (-z_k)^{1/n}$, where the principal n -th root is taken at each step. A naive analysis based on an "unwrapped" angle incorrectly predicts convergence to a single fixed point at $e^{i\pi/(n-1)}$. We demonstrate that this model is flawed due to the discontinuity (branch cut) inherent in the principal root function. The true dynamics, governed by a piecewise angular map, cause the system to converge not to a fixed point, but to a stable 2-cycle between the points $e^{i\pi/(n+1)}$ and $e^{-i\pi/(n+1)}$. We rigorously prove the existence and stability of this limit cycle and provide visualizations of its behavior, showcasing a beautiful example of how discontinuities give rise to more complex periodic attractors.

1 Introduction

The study of sequences generated by iterated functions in the complex plane often reveals intricate and beautiful structures. A canonical example is the sequence $z_{k+1} = z_k^{1/n}$, where the principal n -th root is chosen at each step. For any initial value $z_0 \in \mathbb{C} \setminus \{0\}$, this sequence robustly converges to the fixed point 1.

This paper investigates a subtle modification to this recurrence, where the root is taken of the *negative* of the previous term:

$$z_{k+1} = (-z_k)^{1/n}. \tag{1}$$

This multiplication by -1 corresponds to a π radian phase shift. A simple analysis might suggest this merely shifts the attractor. However, the crucial detail is the behavior of the ****principal root****, which is defined for arguments in the interval $(-\pi, \pi]$. The discontinuity along the negative real axis—the branch cut—is not a mathematical technicality but the dominant feature of the system's dynamics. We will show that this feature prevents convergence to a single fixed point and instead gives rise to a stable, period-2 limit cycle.

2 Convergence of the Modulus

First, we analyze the evolution of the modulus, which is independent of the branch cut's complexities. Let $z_k = r_k e^{i\theta_k}$ be the polar representation. The modulus of the next term, r_{k+1} , is:

$$r_{k+1} = |z_{k+1}| = |(-z_k)^{1/n}| = |(-1 \cdot z_k)|^{1/n} = (|-1| \cdot |z_k|)^{1/n} = |z_k|^{1/n} = r_k^{1/n}.$$

This defines the simple real-valued recurrence $r_{k+1} = r_k^{1/n}$. For any $r_0 > 0$, the sequence $\{r_k\}$ converges to 1. This proves that the ultimate dynamics of the system unfold on the unit circle, which acts as the global attractor for the modulus. Our main task is to understand the behavior of the argument on this circle.

3 Angular Dynamics: A Tale of Two Maps

The long-term behavior is governed entirely by the argument θ_k . The manner in which we model the angle is critical.

3.1 The Idealized (but Flawed) Unwrapped Map

If one ignores the principal root's definition and treats the angle as a continuous real variable (an "unwrapped" angle), the map is simple. The term $-z_k$ has its angle shifted by π , giving the recurrence:

$$\theta_{k+1} = \frac{\theta_k + \pi}{n}. \quad (2)$$

This is a linear contraction map. Its unique fixed point θ^* is found by setting $\theta_{k+1} = \theta_k = \theta^*$:

$$n\theta^* = \theta^* + \pi \implies (n-1)\theta^* = \pi \implies \theta^* = \frac{\pi}{n-1}.$$

This model predicts a single "sliding attractor" at $e^{i\pi/(n-1)}$. While mathematically valid for this specific map, this map **does not represent** the principal root iteration.

3.2 The True Map of the Principal Root

The principal root requires the final argument to be in the interval $(-\pi, \pi]$. Let's derive the correct map. The argument of $-z_k$ is $\theta_k + \pi$. We must map this back to $(-\pi, \pi]$ before dividing by n .

- If $\theta_k \in (-\pi, 0]$, then $\theta_k + \pi \in (0, \pi]$. This is already in the principal range.
- If $\theta_k \in (0, \pi]$, then $\theta_k + \pi \in (\pi, 2\pi]$. To bring this into the principal range, we must subtract 2π , giving an argument of $\theta_k - \pi$.

Thus, the true angular recurrence relation is a piecewise function:

$$\theta_{k+1} = \frac{\text{Arg}(-z_k)}{n} = \begin{cases} (\theta_k + \pi)/n & \text{if } \theta_k \in (-\pi, 0] \\ (\theta_k - \pi)/n & \text{if } \theta_k \in (0, \pi] \end{cases} \quad (3)$$

This discontinuity at $\theta_k = 0$ is the source of the system's rich behavior. A single fixed point cannot exist, as solving $\theta^* = (\theta^* \pm \pi)/n$ yields a value that violates the condition required to use that piece of the function.

4 The Stable 2-Cycle Attractor

We now prove that the system converges to a period-2 limit cycle.

4.1 Proof of the 2-Cycle

Let's hypothesize a 2-cycle between a positive angle θ_A and a negative angle θ_B . Starting with $\theta_A > 0$, the next term θ_B must be generated by the second case of (3):

$$\theta_B = \frac{\theta_A - \pi}{n}. \quad (4)$$

Now starting with $\theta_B \leq 0$, the next term must be θ_A , generated by the first case:

$$\theta_A = \frac{\theta_B + \pi}{n}. \quad (5)$$

We now have a system of two linear equations. Substituting (4) into (5):

$$\begin{aligned} \theta_A &= \frac{1}{n} \left(\frac{\theta_A - \pi}{n} + \pi \right) \\ n\theta_A &= \frac{\theta_A - \pi}{n} + \pi \\ n^2\theta_A &= \theta_A - \pi + n\pi \\ (n^2 - 1)\theta_A &= (n - 1)\pi \\ (n - 1)(n + 1)\theta_A &= (n - 1)\pi \end{aligned}$$

For $n \geq 2$, we can divide by $(n - 1)$ to find the unique solution for θ_A :

$$\theta_A = \frac{\pi}{n + 1}.$$

Substituting this back into (5) gives $\theta_B = n\theta_A - \pi = \frac{n\pi}{n+1} - \pi = \frac{n\pi - (n+1)\pi}{n+1} = -\frac{\pi}{n+1}$. Thus, the system converges to the two-point set $\{e^{i\pi/(n+1)}, e^{-i\pi/(n+1)}\}$.

4.2 Behavior for varying n

As n increases, the two points of the cycle, with angles $\pm\frac{\pi}{n+1}$, move symmetrically towards the positive real axis, eventually converging to 1 as $n \rightarrow \infty$.

5 Conclusion

The recurrence $z_{k+1} = (-z_k)^{1/n}$ is a compelling case study in complex dynamics, offering insights that are deeper than they first appear.

- **Primacy of the Branch Cut:** The analysis demonstrates that the branch cut of the principal root is not a minor detail but the defining feature of the system. It creates a discontinuity that prevents convergence to a single fixed point.
- **Emergence of Periodic Orbits:** Instead of simple convergence, the system settles into a stable period-2 limit cycle. This is a classic example of how even simple iterative functions can produce non-trivial periodic behavior in the complex plane.

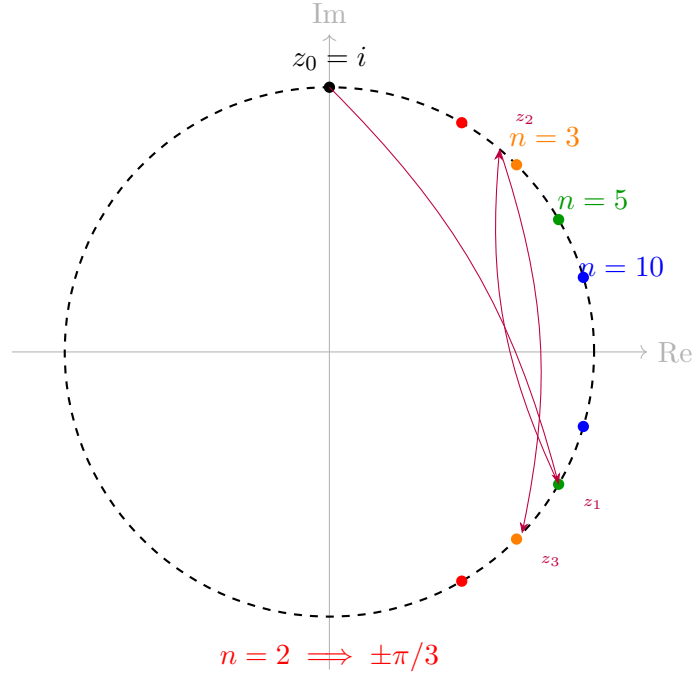


Figure 1: The stable 2-cycle attractors for $z_{k+1} = (-z_k)^{1/n}$ on the complex unit circle. Due to the branch cut of the principal root, the system settles into a periodic orbit between two points with arguments $\theta^* = \pm\pi/(n+1)$. As n increases, this pair of points slides symmetrically towards 1. An example convergence path for $n = 3$ starting at $z_0 = i$ is shown in purple, demonstrating the zig-zagging approach to the 2-cycle at $\pm\pi/4$.

- **Correctness vs. Simplicity:** The contrast between the simple, intuitive (but incorrect) unwrapped model and the piecewise, correct model highlights the need for mathematical rigor when dealing with multi-valued functions.
- **Elegant Geometric Interpretation:** The derived formula for the cycle's arguments, $\theta^* = \pm\pi/(n+1)$, is simple and provides a clear geometric picture. The two attractor points slide symmetrically along the unit circle towards 1 as n increases, a beautiful illustration of how a system's equilibrium state changes with its parameters.

In conclusion, the seemingly simple introduction of a negation transforms the dynamics of iterated roots from convergence to a fixed point to convergence to a stable 2-cycle. This result serves as a powerful and visually rich testament to the complexity and beauty that arise from the fundamental properties of functions in the complex plane.